PANIMALAR ENGINEERING COLLEGE

Department of Artificial Intelligence and Data Science

23ES1511 - TECHNICAL SKILL PRACTICES IV

1. Which of the following is a valid Java variable declaration?

A) `int 1num;`
B) `int num1;`
C) `int num-1;`
D) `int class;`

Answer: B) **`int num1;`**

2. Which data type is used to store a single character in Java?

A) `String`
B) `char`
C) `character`
D) `varchar`

Answer: B) **`char`**

3. Which operator is used for logical AND in Java?

A) `&`
B) `&&`
C) `||`
D) `!&`

Answer: B) **`&&`**

4. Which statement is used to execute a block of code when a condition is true?

A) `switch`
B) `if`
C) `for`
D) `break`

Answer: B) **`if`**

5. Which statement is suitable for selecting one option from multiple fixed values?

A) `if`
B) `for`
C) `switch`
D) `while`

Answer: C) **`switch`**

6. Which keyword is used to add an element to a collection such as `ArrayList`?

A) `insert()`
B) `append()`
C) `add()`
D) `put()`

Answer: C) **`add()`**

7. Which method removes an element from an `ArrayList`?
A) `delete()`
B) `remove()`
C) `discard()`
D) `erase()`

Answer: B) `remove()`

8. Which collection stores data in key-value pairs?
A) `ArrayList`
B) `HashSet`
C) `HashMap`
D) `Stack`

Answer: C) `HashMap`

9. What is the index of the first element in a Java array?
A) 0
B) 1
C) -1
D) 2

Answer: A) 0

10. What is the output of the following code?

```
int[] a = {10, 20, 30, 40};
System.out.println(a[2]);
```

A) 10
B) 20
C) 30
D) 40

Answer: C) 30

11. Which class is commonly used to perform various operations on strings in Java?
A) `String`
B) `StringClass`
C) `CharacterString`
D) `Text`

Answer: A) `String`

12. Which method returns the number of characters in a String?
A) `size()`
B) `length()`
C) `count()`
D) `getSize()`

Answer: B) `length()`

13. Which of the following correctly demonstrates a lambda expression in Java?
A) `x -> x * 2`

B) `x => x * 2`
C) `lambda x: x * 2`
D) `x := x * 2`

Answer: A) `x -> x * 2`

14. Which package contains classes for handling date and time in modern Java?
A) `java.time`
B) `java.date`
C) `java.datetime`
D) `java.calendar.time`

Answer: A) `java.time`

15. Which class is used to represent a date without time in Java?
A) `DateOnly`
B) `LocalDate`
C) `DateTime`
D) `CalendarDate`

Answer: B) `LocalDate`

16. Which algorithm repeatedly compares adjacent elements and swaps them if they are in the wrong order?
A) Insertion Sort
B) Selection Sort
C) Bubble Sort
D) Merge Sort

Answer: C) Bubble Sort

17. In insertion sort, an element is placed into:
A) The last position
B) Its appropriate position in the sorted portion
C) The first position always
D) A random position

Answer: B) Its appropriate position in the sorted portion

18. Which class is commonly used to read text from a file in Java?
A) `FileReader`
B) `FileWriter`
C) `FilePrinter`
D) `TextReader`

Answer: A) `FileReader`

19. Which keyword is used to handle an exception in Java?
A) `error`
B) `catch`
C) `exception`
D) `handle`

Answer: B) `catch`

20. Which command-line array stores the arguments passed to the `main()` method?
A) `String[] args`
B) `int[] args`
C) `String args()`
D) `Command[] args`

Answer: A) `String[] args`

21. Which statistical measure represents the arithmetic average of a dataset?
A) Median
B) Mode
C) Mean
D) Variance

Answer: C) Mean

22. Which statistical measure represents the middle value after arranging data in ascending order?
A) Mean
B) Median
C) Mode
D) Standard deviation

Answer: B) Median

23. Which statistical measure represents the most frequently occurring value?
A) Mean
B) Median
C) Mode
D) Variance

Answer: C) Mode

24. Standard deviation is primarily a measure of:
A) Central tendency
B) Data dispersion
C) Data frequency
D) Data classification

Answer: B) Data dispersion

25. Which keyword is used to create a package in Java?
A) `package`
B) `import`
C) `module`
D) `namespace`

Answer: A) `package`

26. In Java, which keyword is used to create an object?
A) `object`
B) `create`

C) `new`
D) `instance`

Answer: C) `new`

27. Which machine learning algorithm is based on Bayes' theorem?
A) K-Means
B) Naïve Bayes
C) ID3
D) Linear Regression

Answer: B) Naïve Bayes

28. K-Means clustering is primarily used for:
A) Classification
B) Clustering
C) Regression
D) Association rule mining

Answer: B) Clustering

29. Which criterion is commonly used by the ID3 algorithm to select the best attribute for splitting?
A) Gini Index
B) Information Gain
C) Euclidean Distance
D) Mean Squared Error

Answer: B) Information Gain

30. Which type of chart is most appropriate for showing the relationship between two numerical variables in regression analysis?
A) Pie chart
B) Scatter plot
C) Bar chart
D) Histogram

Answer: B) Scatter plot